SAURABH BELGAONKAR

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EDUCATION

TEXAS A&M UNIVERSITY 2023 - 2027
Ph.D. Mechanical Engineering GPA: 4.0/4.0
Advisor: Dr.Sivakumar Rathinam

IISC BANGALORE Oct 2020 – Jul 2022
M.Tech Mechanical Engineering
GPA: 9.2/10

NIT WARANGAL Aug 2015 – May 2019
B.Tech Mechanical Engineering
GPA: 8.92/10

SKILLS

Programming & Systems
C++, Python, ROS2, Linux, Git, SQL, OpenCV, Docker, Databricks, Spark

Perception & ML
Point clouds, PCL, OPEN3D, PointPillars, 3D object detection, Deep Learning, Tensorflow, PyTorch, CUDA, TensorRT, MuJoCO, 3D geometry

Robotics & Sensing
Camera–LiDAR calibration, sensor fusion, multi-object tracking, segmentation

Optimization
Linear Algebra, Gurobi, MIP, convex optimization, dynamic programming, LKH

SELECTED COURSES

Convex Optimization, Reinforcement Learning, Intelligent Systems, Control Systems, Multi-modal Computer Vision

PUBLICATIONS

- J. Keshavan, **S. Belgaonkar**, "Adaptive Super-Twisting Control of a First-Order Sliding Mode with an Output Constraint," *IEEE Access*.
- **S. Belgaonkar** et al., "A Mixed-Integer Convex Optimization based approach for Hybrid UAV Path Planning in Noise-Restricted Zones," *IEEE Transactions on Aerospace and Electronic Systems*.
- **S. Belgaonkar** et al., "Pareto Front Approximation for the Biobjective Steiner Traveling Salesman Problem," *IROS*.

TECHNICAL STRENGTHS

- Real-time **C++/ROS2** autonomous driving software
- Camera–LiDAR perception, sensor fusion, and **SLAM** pipelines
- **MIP** and convex optimization with **Gurobi**
- ML model training, edge deployment, and inference acceleration

ACHIEVEMENTS

- Led an autonomous car competition team of 35 to consecutive 2nd Place finishes Autodrive Challenge II(2024,2025).
- Awarded the **S. V. Sastry Memorial Gold Medal**, IISc Bangalore, 2022.

PROFESSIONAL EXPERIENCE

CAPTAIN; PERCEPTION/COMPUTER VISION LEAD Oct 2023 – Present
GM SAE AutoDrive Challenge
- Enabled reliable object detection and tracking under **adverse conditions** (fog) using a **C++/ROS2** multi-modal perception stack with **LiDAR, radar, and cameras**.
- Led real-time perception deployment for **RF-DETR**, **U-Net**, **PointPillars**, and LiDAR clustering, increasing full-stack throughput by approximately **50%** to **15 Hz**.
- Developed navigation logic with the **Nav2 Behavior Tree** framework, using object detections to trigger modular obstacle avoidance and scene-specific behaviors.
- Enabled autonomous operation in **GPS-denied** environments by integrating **IMU, GNSS, cameras, and LiDAR** for pose estimation with **LiDAR SLAM**.

DATA SCIENTIST Aug 2022 – Jul 2023
Atlas Copco, India
- Built machine-health monitoring pipelines with **CNNs**, **GANs**, signal processing, and **Databricks** enabling early fault prediction that reduced unexpected downtime.
- Built a vibration-based compressor speed estimator using 1D **CNN** features, and **LightGBM**, improving accuracy from **70%** to **99.5%** using limited labeled data.

SELECTED PROJECTS

REAL-TIME OBJECT DETECTION DEPLOYMENT Feb 2024 – Present
GM SAE AutoDrive Challenge
- Trained transformer-based **RF-DETR** obstacle detector for **14 driving-scene classes** including barrels, pedestrians, vehicles, and deer, achieving **89.3 mAP@0.5**.
- Accelerated multi-camera **RF-DETR** inference in **C++** using **TensorRT**, parallel asynchronous stages, and batched GPU inference, reducing runtime by **60%**.
- Deployed calibrated **INT8 quantization** for **RF-DETR**, reducing detector latency by an additional **35%** with only a **0.91 mAP-point** validation drop.

LIDAR PERCEPTION AND MULTI-SENSOR FUSION Feb 2024 – Present
GM SAE AutoDrive Challenge
- Improved obstacle extraction and **pose estimation** using **KD-tree**-based spatial indexing, ground segmentation, and **curved-voxel clustering**, achieving **25%** higher accuracy and **1.8**× faster runtime than **DBSCAN** clustering.
- Deployed **PointPillars**-based LiDAR detector for pedestrians, vehicles, barrels, and animals classification with **Kalman filtering** for stable **multi-object tracking**.
- Fused camera and LiDAR detections using projected-box **IoU** and **Hungarian** assignment, improving adverse-condition detection by **32%**.

FEATURE-LEVEL CAMERA–LIDAR FUSION Apr 2025 – Present
GM SAE AutoDrive Challenge
- Built **YOLOv11**-based **Siamese** RGB–LiDAR detector with feature-level fusion of RGB and projected LiDAR depth inputs for robust **multi-modal** object detection.
- Evaluated against a camera-only baseline, improving detection by approximately **45%** under lighting variation, fog, and glare while retaining real-time inference.

OPTIMIZATION-BASED ENERGY-AWARE ROUTING Feb 2024 – Jan 2025
Autonomous Systems Lab, Texas A&M University
- Formulated continuous-space path and energy planning for **hybrid UAVs** under **battery**, & electric-only **noise-restricted-zone** constraints and solved it as a **mixed-integer convex program (MICP)**, achieving up to **10**× speedup over the baseline.
- Extended the MICP framework to **TSP**, discretizing state-of-charge to convert the problem into **GTSP** and solving with **LKH** for **5**× improvement over baseline.
- Developed a load-dependent energy model and **branch-and-cut MILP** solver for **Load dependent TSP**, solving 50-target instances to optimality in under 1 minute.

AUDIO-BASED LUBRICANT LEVEL ESTIMATION Mar 2023 – Sep 2023
Atlas Copco, India
- Built a **CNN** audio pipeline for lubricant-level estimation on large-scale data using **Mel spectrogram** features, **SMOTE**, and augmentation.
- Built a **Databricks** training and deployment workflow with **Python**, **PyTorch**, and **Spark**, achieving an $\mathbf{R^2}$ **score of 0.92** in production inference.